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Revolutionizing Sidetrack Planning: An Innovative Automated Workflow for Opportunity Generation and Scheduling

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When operating mature or late life oil and gas fields the objective is focused on using the existing infrastructure to economically recover as much of the in-place oil and gas as possible. For offshore fields at the later stages there can often be insufficient reserves to justify any new infrastructure in the asset. The primary approach to enhancing production now centers on interventions in existing wells to sustain output from current reservoir locations. As these wells become less economically viable, the strategy shifts toward utilizing depleted wells to access remaining reserves within the reservoir by drilling sidetracks into identified hydrocarbon pockets.

Given that the field will be heading towards the end of its life, there needs to be a very strong focus on the investment returns to keep the whole operation economic. In the North Sea there are many large assets that have been operating for many decades. The fields have been developed with fixed platforms using platform wells, floating platforms where the subsea wells can be accessed by the platform's drilling rig and more remote wells that can only be accessed by a mobile rig. When looking at drilling new reservoir targets from the existing field platforms, consideration must be given to viability of each sidetrack, the size of the recoverable volume in the reservoir target and type of well design to reach the reservoir target. The risk elements considered in this process are associated with each sidetrack plan including volume risk of the reservoir target, plug and abandonment risks for slot recovery and wellpath design risk.

To mitigate the risks for the development program, a portfolio approach can be considered to balance the return and the risks associated with the program to successfully extend the production life of the asset. In many old assets the largest remaining opportunities may be on the periphery of the reservoir, which may require the longest reach wells and may be riskier, whereas smaller pockets of bypassed oil may be closer to the existing wells in the middle of the field and can be reached with shorter sidetracks from deeper depths in the mother bore.

Introduction

When operating mature or late life oil and gas fields the objective is focused on using the existing infrastructure to economically recover as much of the in-place oil and gas as possible. For offshore fields at the later stages there can often be insufficient reserves to justify any new infrastructure in the asset. The primary approach to enhancing production now centers on interventions in existing wells to sustain output from current reservoir locations. As these wells become less economically viable, the strategy shifts toward utilizing depleted wells to access remaining reserves within the reservoir by drilling sidetracks into identified hydrocarbon pockets.

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To mitigate the risks for the development program, a portfolio approach can be considered to balance the return and the risks associated with the program to successfully extend the production life of the asset. In many old assets the largest remaining opportunities may be on the periphery of the reservoir, which may require the longest reach wells and may be riskier, whereas smaller pockets of bypassed oil may be closer to the existing wells in the middle of the field and can be reached with shorter sidetracks from deeper depths in the mother bore.

Many of these fields have 10's of existing wells and may have 10's of potential reservoir targets that can be sidetracked from different kick-off points in the mother bore. Theoretically this leads to 100's of potential sidetracks opportunities. In addition, there is an added complication that the existing well bores will become available at different times as their reserves deplete. The new reservoir targets may be close enough to be reached from several of the existing wells. The selection of one wellbore versus another may lead to a result that rules out other potential reservoir targets and an attractive opportunity may be lost.

To deliver a comprehensive analysis requires significant collaboration between the reservoir engineer and the well engineers. There are several new workflows for the identification of remaining reserves. The well engineers must assess the current state of the existing wells to check for their suitability for re-use. To assess all the reservoir target to well side-track combinations is a daunting task and may not be possible without innovative digital solutions.

This paper presents the results of a collaborative innovation project with Petoro AS to develop a solution to address the challenge described above, and the results of the initial testing on a mature asset in the Norwegian Continental Shelf.

Method

This section provides a comprehensive overview of the key components that constitute the end-to-end digital solution known as Drilling Portfolio IQ. We begin with a high-level overview to establish the overall structure and purpose of the workflow. Following that, each component is examined individually, with an in-depth discussion of its role, functionality, and integration within the broader system.

This solution was built on a cloud-based platform that enables users to develop and integrate custom engines to support and accelerate decision-making across various stages of field development planning. As illustrated in [Figure 1](#), it is composed of three main components: the Sidetrack Generation Workflow, the Opportunity Register, and the Sidetrack Scheduling Workflow.

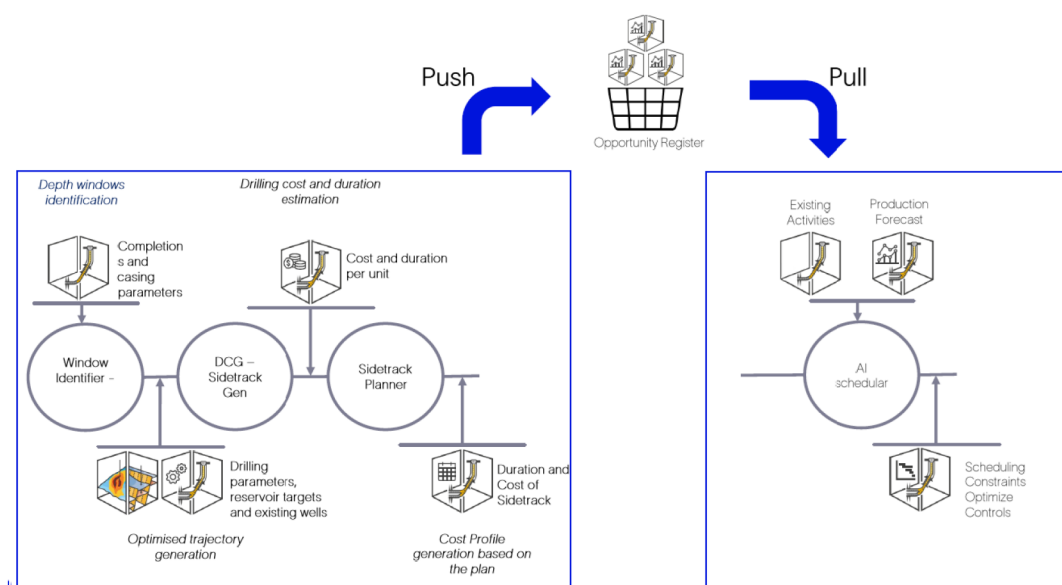


Figure 1—Drilling portfolio IQ overview – This figure illustrates the three core components of the Drilling Portfolio IQ solution: the Sidetrack Generator, the Opportunity Register, and the Sidetrack Opportunity Scheduling Workflow.

The user journey begins with the Sidetrack Generation Workflow. Here, relevant data for existing wells and predefined targets are first loaded into the system. The user then defines combinations of wells and targets that can potentially be connected through sidetracks. Each well-target pair is treated as a unique scenario within the platform. These scenarios are individually simulated, and the results are systematically recorded. Once all scenarios have been executed, the user evaluates the simulation results to identify the most promising sidetrack candidates. These selected scenarios are then promoted and published to the Opportunity Register.

The promoted sidetracks, along with their associated data, are subsequently fed into a dedicated study that is linked to the Sidetrack Scheduling Workflow. In this stage, users can run multiple scheduling scenarios under varying constraints. The scheduling engine using an optimization algorithm determines the optimal combination of sidetracks to be drilled within a specified timeframe and based on a user defined technical constraint.

Sidetrack generation workflow

Figure 2 illustrates the sidetrack generation workflow on the cloud-based platform used for computational processing. In general, a workflow consists of interconnected modules—represented as circles—that serve as compute engines. These engines are linked by cards, which act as data sources, either as user-defined inputs or as outputs produced by the compute engines.

In this specific workflow, three computes are connected sequentially to enable the generation and evaluation of sidetrack trajectories. The Sidetrack Window Identifier first determines a suitable kickoff window for initiating the sidetrack from the existing wellbore. Next, the Sidetrack Generator designs the trajectory based on user-defined drilling constraints, such as dogleg severity (DLS), inclination and wellbore total length. Finally, the Sidetrack Planner computes key technical indicators—such as risk, wellbore geometry and cost—which are used to assess the feasibility and performance of each generated sidetrack option.

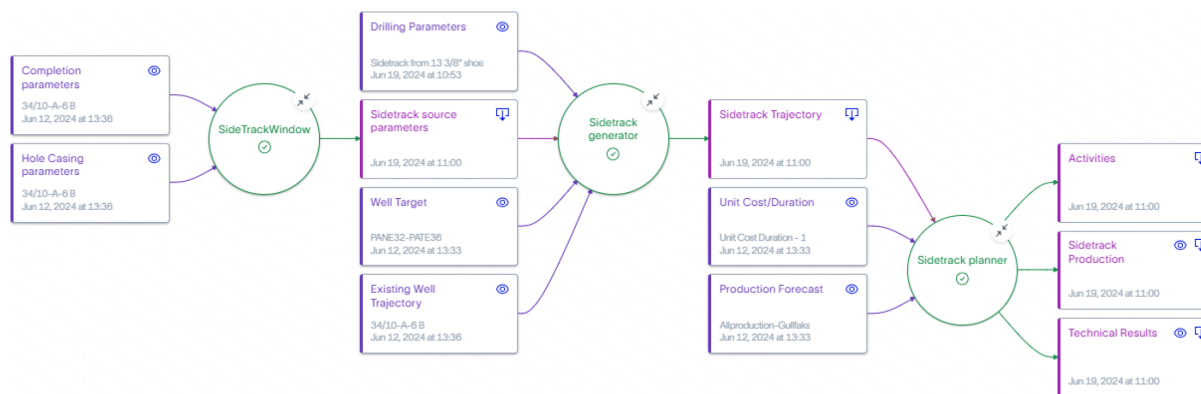


Figure 2—Sidetrack generator workflow inputs and computations – This figure presents the essential data inputs and resulting outputs required to execute the Sidetrack Generator workflow. The process comprises three key computational modules: the Sidetrack Window Identifier, the Sidetrack Generator, and the Sidetrack Planner.

Sidetrack Window identifier. The Sidetrack Window Identifier computation uses algorithm to identify the depths of potential kick-off windows and drift diameters for each window in the existing wells based on two inputs: completion and casing input parameters.

Inputs

There are two main input data sources that are required for the sidetrack window identifier, namely the completion parameters, and the hole casing parameters as shown in [Figure 2](#).

- Completion parameters data source presents depth information for different components of a completion string for existing wells.
- Hole casing parameters data source presents depth information for different casing sections of the existing well, such as conductor, surface casing, and intermediate casing.

Output

- Sidetrack source parameters data source present depth information about each identified kickoff window along with the drift diameter for each window. The drift diameter marks the first casing size of the sidetrack well.

Compute

The compute first performs data cleaning to eliminate erroneous or inconsistent entries. It then processes top-of-cement (TOC) data for the various casing strings to identify intervals where multiple casings are cemented together. This allows the algorithm to differentiate between sections where a single casing is cemented to the formation and sections where overlapping cemented casings exist. The underlying assumption is that milling a window through a single cemented casing is significantly easier and more efficient than milling through multiple cemented layers.

Additionally, the algorithm incorporates well completion data to exclude zones that are inaccessible to milling operations due to installed hardware. For instance, if a permanent packer is present, the algorithm automatically filters out all intervals below the packer, as milling through such equipment is typically complex, time-consuming, and operationally risky.

Sidetrack Generator. The Sidetrack Generator compute uses an existing well trajectory, a specified well target, sidetrack kick-off windows, and a list of drilling constraints to generate an optimized sidetrack trajectory.

Inputs

There are four main input data sources that are required for the sidetrack generator, namely the sidetrack source parameters, drilling parameters, well target and existing well trajectory as shown in [Figure 2](#).

- Sidetrack source parameters data source is generated by the sidetrack window identifier. The workflow is designed to be modular, allowing users the flexibility to run the sidetrack generator independently. In such cases, this data source can also be manually defined by the user.
- Drilling parameters data source contains the drilling input data and constraints required for the well design process. Key parameters include:
 - Dogleg Severity (DLS): The maximum allowable curvature of the wellbore.
 - Horizontal Step-out: The maximum horizontal drilling distance that the rig can accommodate.
 - Kick-off Window: The user-defined depth interval for initiating the sidetrack, which can be specified as shallow, medium, or deep.
 - Number of Casing Strings: Specifies how many casings are to be used in the sidetrack.
 - Final Casing Size: The diameter of the final casing string in the well.
- Well target data source defines the trajectory points of the intended reservoir target. Each target is represented by two spatial coordinates within the reservoir model.
- Existing well trajectory data source contains the trajectory data of the existing (mother) wellbore from where the sidetrack will be initiated.

Output

- Sidetrack trajectory data source contains the trajectory data of the generated sidetrack along with the number of casings involved.

Compute

The compute designs each sidetrack trajectory using the Minimum Curvature Method as described in (Sawaryn & Thorogood, 2003). It generates the shortest possible trajectory connecting the kick-off point to the target while honouring key geometric and operational constraints, including the dogleg severity at both the kick-off and target points, the horizontal step-out between them, and the maximum allowable inclination of the tangent section. The compute also ensures to kick off with a smooth trajectory with the same azimuth of the mother wellbore. Additionally, the algorithm determines the number of casing strings required for each sidetrack by applying a default casing design. It is calculated based on the user-defined drift diameter and the diameter of the last casing section.

Sidetrack planner. The Sidetrack planner compute calculates the technical indicators, the drilling cost and duration and the production profile of each sidetrack.

Inputs

There are three main input data sources that are required for the sidetrack planner, namely the sidetrack trajectory, unit cost/duration/risk, well and production forecast as shown in [Figure 2](#).

- Sidetrack trajectory data source is generated as an output of the sidetrack generator.
- Unit cost/duration data source contains multiplier parameters used to estimate the cost, duration, and risk associated with the sidetrack trajectory.
- Production forecast data source provides the unconstrained production forecast for all the targets involved in the study

Output

- Technical results data source includes all technical indicators utilized by the user to evaluate sidetracks within the sidetrack generation study. These indicators include cumulative oil production, cumulative gas production, cost, duration, risk and length of the generated sidetrack.

- Sidetrack production data source includes the unconstrained production forecast for the target of the specific sidetrack being evaluated within a given scenario
- Activities data source includes the cost, duration, and risk values associated with the drilling operations of the generated sidetrack.

Compute

The compute performs simple calculations to estimate the key technical indicators of a sidetrack, including cost, duration, and risk. To estimate cost, the user provides a unit cost per day and defines an approximate drilling duration for three sidetrack categories: deep, intermediate, and shallow. These categories are determined based on the number of casings used in the sidetrack. Specifically, a sidetrack with one casing is classified as deep, with two casings as intermediate, and with three or more casings as shallow. Using this classification, the compute selects the appropriate duration and multiplies it by the unit cost per day to determine the total cost.

In addition to cost and duration, the compute also calculates the sidetrack's risk. Risk is evaluated across three categories: inclination, casing design, and sidetrack length. For inclination and length, the user defines minimum and maximum bounds and assigns risk factors to each range—low, intermediate, and high. For casing design, a risk factor is assigned based on the number of casings. The compute also determines which range each characteristic falls into and selects the corresponding risk factor. These individual risk values are then summed to produce the final risk score for the sidetrack.

Furthermore, the compute retrieves the corresponding production forecast for the sidetrack under evaluation and populates it into the Sidetrack Production data source. It is important to note that both the Activities and Sidetrack Production data are carried over from the Opportunity Register into the Sidetrack Scheduling Study, as will be discussed in the following sections.

Opportunity register

The Opportunity Register captures investment decisions across various phases of the Exploration and Production (E&P) lifecycle, including Exploration, Capital Project Development, and Asset Operations. Each opportunity is defined by its associated subsurface resources and preferred infrastructure—such as facilities, subsea platforms, and other relevant components—and is enriched with metadata that describes its geospatial location and business context.

To ensure the protection of sensitive information, the register incorporates robust data access controls.

In addition to contextual metadata, opportunities include technical data such as production forecasts, cost estimates, cash flow projections, and economic indicators. These are derived from multiple scenarios evaluated during FDPlan studies at different project milestones, establishing an auditable lineage throughout the maturation journey.

The register also serves as a repository of domain expertise, capturing insights from specialists to support informed decision-making across the lifecycle of each opportunity.

The project leverages this feature to conduct sidetrack analysis for new subsurface targets across multiple studies, each treated as a distinct scenario. Scenarios identified as potential sidetrack candidates are promoted to the Opportunity Register (Figure 3), where their technical data evaluated within a dedicated study. A new study is then initiated to perform scheduling analysis on these opportunities using established scheduling workflows. The Opportunity Register enables users to filter and prioritize sidetrack opportunities, facilitating their selection for subsequent workflows such as scheduling sidetracks to optimize production.

Opportunity name	Duration	Gas Production	Length	Oil Equivalent Production	Oil Production	Risk
Sidetrack_5	104.23	17 44.48	2963.14	2.09	0.35	18
Sidetrack_1	137.03	40.19	5160.91	0.33	0.29	20
Sidetrack_6	140.76	31.44	5410.79	0.21	0.17	20
Sidetrack_7	141.59	31.44	5466.49	0.21	0.17	20
Sidetrack_8	140.39	65.72	5386.08	0.23	0.16	20
Sidetrack_4	141.37	48.19	5451.52	0.14	0.09	20

Figure 3—Opportunity Register User Interface: This figure illustrates the user interface for accessing the Opportunity Register feature and its associated records. It showcases various attributes and provides filtering capabilities to enhance data navigation and analysis.

The sidetrack scheduling workflow

The Sidetrack Scheduling workflow contains a single compute module—the Sidetrack Scheduler—which is responsible for optimizing the drilling schedule of the sidetracks (see Figure 4).



Figure 4—Sidetrack scheduler workflow inputs and computations – This figure presents the essential data inputs and resulting outputs required to execute the Sidetrack scheduler workflow. The process comprises a key computational compute: the Sidetrack scheduler.

Sidetrack scheduler. The Sidetrack scheduler is a model-based optimization engine utilizing systematic and local search. The model is based on a custom[AH1] task model and a PDDL (Planning Domain Definition Language) 2.1 (Fox & Long, 2003) (Long et al., 2023) model which allows for the definition of complex relations (e.g., mutual exclusion of drilling targets), numeric constraints (e.g., Platform capacity), and numeric optimization criteria (e.g., fluid production and risk). The model-based approach also ensures the interpretability of the results and precise control over the data and the optimization process.

Inputs

The inputs consist of two main categories. The first category includes the results of the sidetrack generation workflow, which are mainly the sidetracks cost, duration, risk (which are represented by the "activities" datasource) and the production profile for each sidetrack (which is represented by the Sidetrack production data source). Both data sources are coming from the opportunity register.

The second category includes the constraints and input parameters that are loaded manually by the user. These inputs include:

- Host production data sources include the current production of the existing wells that will be used for sidetrack. This is important as any host well which is used for a sidetrack will lose its current production and this has to be accounted for in the objective function. The
- Scheduling constraints data source includes the facility capacity constraint, the maxim allowable risk and maximum allowable yearly budget,
- Resource allocation data sources include allocation of resources (rigs) to different platforms if needed.
- Resource availability data sources include the rigs availability period throughout the scheduling timeframe.
- Optimization configuration data source includes the weights of the objective function. The objective function includes 4 components: discounted oil production, discounted gas production, total risk and total cost. User can give different weight to each component based on his/her optimization objectives.

Outputs

The output of the compute consists of the schedule, that is, a selection of sidetracks and an assignment of time and resources for each selected sidetrack and other activities, including resource maintenance. Also, the phased production is reported as a time series data.

Compute

The compute proceeds in three steps. The first step is an analysis of the inputs and construction of internal representation of the model. This step assigns a grounded instance of the PDDL action to each sidetrack and activity and initializes the numeric constraints and variables. The second step is the construction of a feasible solution, the greedy score-aware allocation. Iteratively, the scheduler finds the best activity or sidetrack to allocate, that is, determine its start time and resource to perform it. All constraints are respected, and the selection is based on the evaluation of the weighted sum of the metrics, the score, according to the configuration. The last step is optional and performs optimization using a local tabu search ([Glover, 1986](#)) with the large neighborhood search heuristic ([Pisinger & Ropke, 2010](#)) possibly improving the solution. The last step is not always necessary, nor possible, due to the complexity of the constraints.

Results

In this section, we present how the Drilling Portfolio IQ solution was applied to optimize the drilling schedule for a mature offshore field in the North Sea. We begin with an overview of the field and the associated challenges, followed by a description of the targets and well slots identification. Next, we present the results of the sidetrack generated within the Sidetrack Generation workflow, and finally, we showcase the different scheduling scenarios that were evaluated.

The field

A mature field in the North Sea was used as a test case to showcase value. The selected field has produced oil and gas for several decades and has a large production platform with more than 40 well slots ([figure 5](#)). Wellheads and production trees are placed over two shafts, accessing the seafloor through the "legs" of the platform. The main platform drilling unit is installed on skid beams and moved between the two main well shafts depending on the drilling schedules.

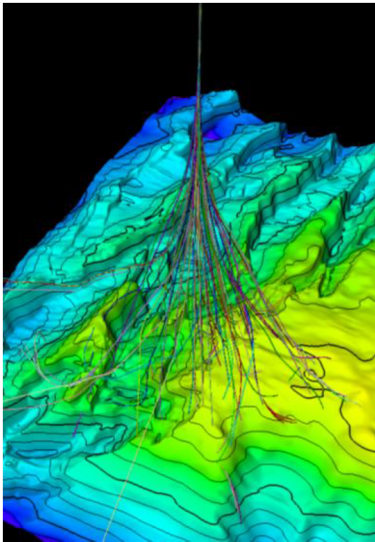


Figure 5—Existing well bores in field study.

The main field is a highly faulted oil field with several reservoir units. Improved recovery measures include horizontal and extended reach wells, new technology within completion and sand control and water alternating gas injection (WAG). The field has large remaining reserves, but target sizes are getting smaller in size and more complex to develop

One of the key challenges to solve, in addition to maturing of new reservoir targets, is to identify the well slots that could be reused as redrilling candidates.

Reservoir targets identification

Target maturation and hunting are critical processes for identifying remaining resources and sustaining the required production levels from the field. targets were identified through detailed analysis of existing well performance, combined with reservoir engineering judgement.

Table 1—Framing of the sidetrack generation workflow. 3 decisions were considered: targets, well slots and sidetrack depth

Decision 1: Targets	Decision 2: Well slots	Decision 3: Sidetrack depth
Target 1	Well #1	Shallow
...	...	Intermediate
...	...	Deep
...	...	
Target 17	...	
	...	
	Well #20	

Engineers focused on areas near high-performing wells and prioritized placement in higher reservoir layers to delay water breakthrough. The production forecast for each target was approximated using the production profile of the nearest existing well. As summarized in Table 1, a total of 17 targets were identified, with estimated production capacities derived from field experience. These target candidates were further categorized based on oil and gas volumes. The well target data source for each target was prepared and loaded in the sidetrack generation workflow study.

Well slot identification

The slot categorization and selection process is important to understand the best way to utilize the starting point for the new well proposals. All wells not in daily production are identified and categorized based on integrity status and current well design. Then each of the identified re-use candidates are categorized with respect to potential kick-off point for a new well path. Kick-off points are defined as shallow, intermediate and deep. For this specific field case the shallow sidetrack depth is from 20" surface casing shoe. The intermediate sidetrack depth from the 13 3/8" intermediate casing and the deep sidetrack point from the 9 5/8" production casing. A total of 20 well slots were selected and categorized as re-use candidates (Table 1). Once wells have been identified, the hole case parameters, the completion parameters and the well trajectory data source of each of the selected wells are prepared and uploaded into the sidetrack generation workflow study.

Sidetrack generation and evaluation

Once the targets and well slots are identified and all relevant data sources are prepared, the user initiates a study and associates the Sidetrack Generation Workflow with it. The study begins with a framing exercise, during which the user defines the key decisions and choices to be explored across different scenarios. As illustrated in Table 1, the framing for the current use case includes three primary decision variables: target location, well slot selection, and sidetrack depth.

This framing exercise results in a total of 1,020 potential scenarios, derived from the combination of 20 wells, 17 targets, and 3 sidetrack window options ($20 \times 17 \times 3$). Each scenario represents a unique sidetrack opportunity to be evaluated within the workflow.

Drilling parameters are configured within the platform to establish the boundary conditions for well design proposals, tailored to the specific kick-off points of the various sidetrack options. The parameters used in this study are detailed in Table 2. Additionally, the unit cost, duration, and risk data sources were developed based on historical drilling experience and operational insights from the field.

Table 2—Drilling parameters that were considered for generating the sidetrack opportunities.

Drilling parameters	Deep sidetrack from 9 5/8" shoe	Intermediate sidetrack from 13 3/8" shoe	Shallow sidetrack from 20" shoe
DLS at kick off point (deg/30m)	4	3.5	3
DLS at target (deg/30m)	3	3	3
Final hole size (inches)	6	6	6
Max inclination (degrees)	90	90	90
Max horizontal step out (m)	3000	4500	6000
Number of casing sections	2	3	4

After all inputs are defined and correctly linked to their corresponding choices, the simulation is initiated. It is important to note that some combinations may fail due to drilling constraints—these are automatically flagged by the platform. The first output consists of the realistic sidetrack opportunities, as shown in Figure 6, where 230 feasible sidetrack opportunities were identified out of the 1,020 theoretical combinations evaluated during the simulation.

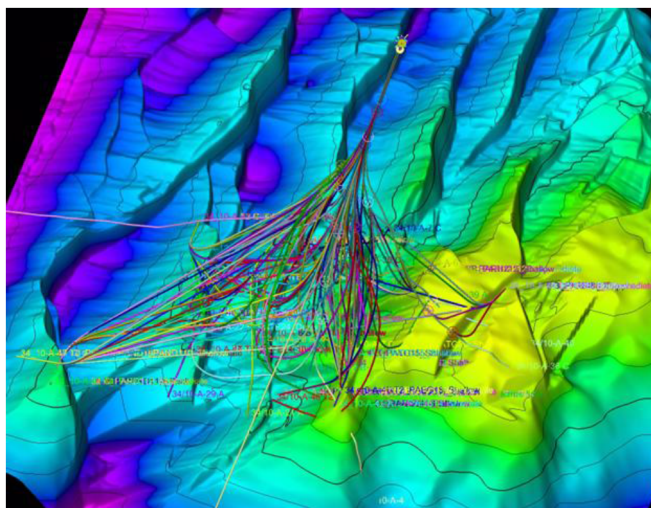


Figure 6—Trajectories of 131 sidetrack opportunities that were generated in the sidetrack generation workflow study.

A detailed evaluation of the proposed well path options was carried out through visual inspection of the well path geometries, complemented by applying cutoffs to each technical indicator. This filtering process reduced the set of opportunities from 230 to 131 high-potential sidetrack options. These selected opportunities, along with their associated activity and production data, were then promoted to the Opportunity Register for use in the scheduling scenarios.

It is important to note that the selected opportunities are not mutually exclusive—some sidetracks originate from the same well or target the same reservoir location. As will be discussed later, the scheduler enforces mutual exclusivity constraints during the scheduling process to ensure that only a unique sidetrack is selected per well or per target

Sidetrack scheduling

After finalizing the list of sidetrack opportunities, we created a new study and associated it with the sidetrack scheduling workflow. 131 sidetrack opportunities were loaded from the opportunity register along with their data. Following that, another framing exercise was conducted—this time focusing solely on a single decision variable: the weighting of the objective function. As we mentioned in the method section, the scheduler optimizes the selection and sequencing of sidetracks based on discounted hydrocarbon production. Note that both oil and gas volumes are measured in barrels of oil equivalent (BOE), and the user can assign different weights to each within the objective function to reflect their relative importance. Three primary weighting strategies were defined, resulting in three distinct scheduling scenarios, as described below.

- Scenario 1: Optimize drilling schedule versus discounted oil and gas production
- Scenario 2: Optimize drilling schedule versus mainly discounted oil production
- Scenario 3: Optimize drilling schedule versus discounted oil and gas production with no high risk wells (risk cutoff at 16)

No constraints were applied on facility capacity, budget or project timeframe in any of the tested scenarios. However, in Scenario 3, a risk constraint was introduced—limiting the total risk score to a maximum of 16—to exclude high-risk opportunities from the scheduling results. The primary objective of this exercise was to observe and compare the optimal set of mutually exclusive opportunities selected by the scheduler under the defined constraints and the relevant objective function. It is worth noting that the user has the flexibility to perform sensitivity analyses on various parameters—such as risk thresholds, cost assumptions, and production capacities—for other use cases or scenarios.

Figures 7, 8, and 9 illustrate the scheduling results for Scenarios 1, 2, and 3, respectively, while Table 3 summarizes the key technical indicators for each scenario.

Table 4—Technical indicators for the schedules of sidetrack opportunities in scenario 1, 2 and 3.

	Number of opportunities	Highest Risk	Average risk	Discounted oil volume (sm ³)	Discounted gas volume (sm ³)	Cumulative cost (million NOK)
Scenario 1	17	18	13.65	1.96	5.33	3862
Scenario 2	17	18	13.65	2.02	5.17	4902
Scenario 3	16	16	12.88	1.92	4.97	3469

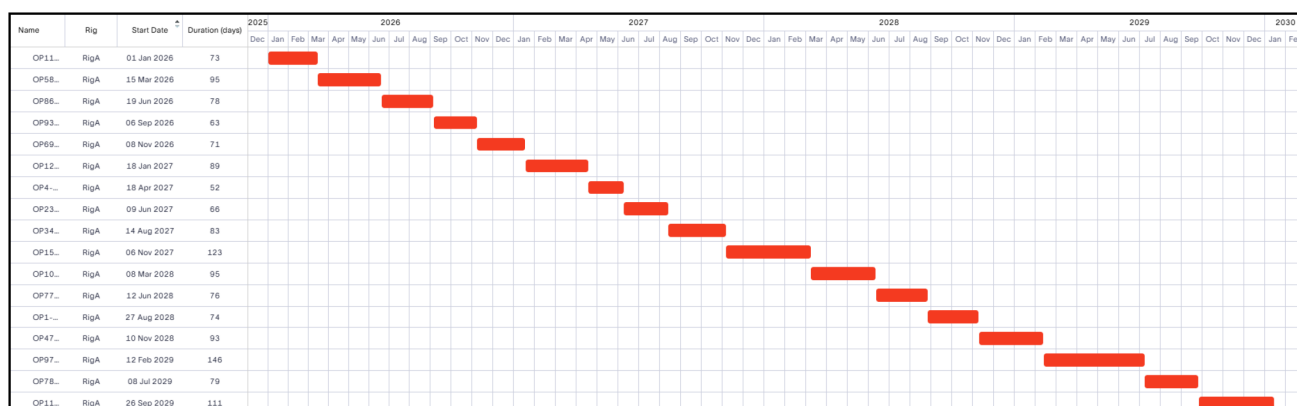


Figure 7—Drilling schedule of scenario 1.

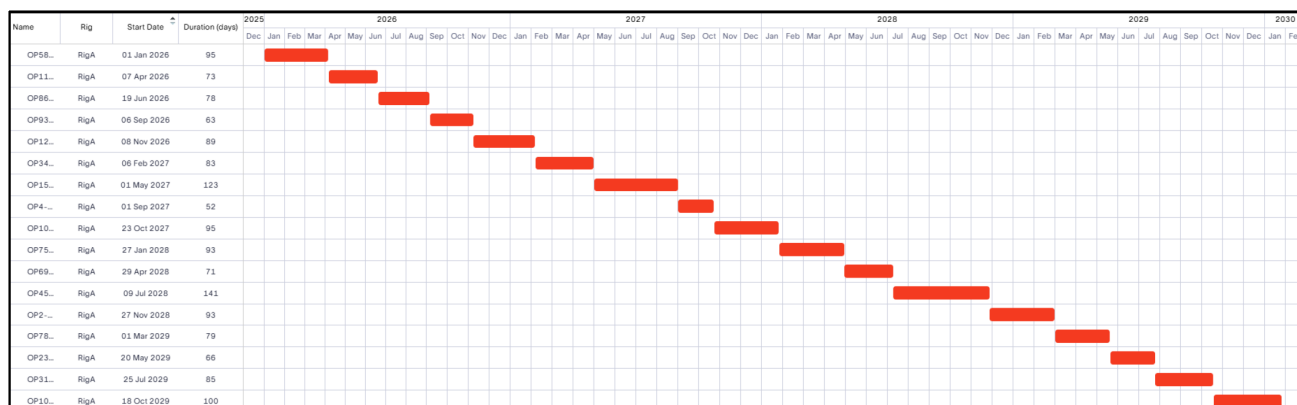


Figure 8—Drilling schedule of scenario 2.

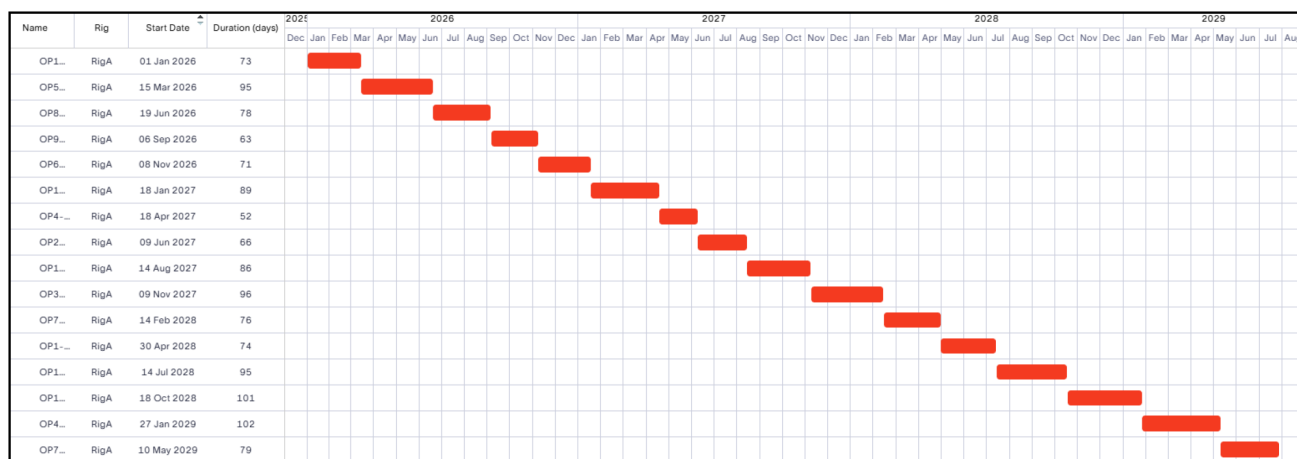


Figure 9—Drilling schedule of scenario 3.

In Scenario 1, the scheduler selected 17 mutually exclusive opportunities, with the first sidetrack starting on January 1, 2026, and the last one concluding on December 15, 2029. Similarly, Scenario 2 also resulted in 17 scheduled opportunities, starting on January 1, 2026, and ending on January 1, 2030. However, in this scenario, the scheduler prioritized oil-driven opportunities, which generally required longer drilling durations. As reflected in the results, the discounted oil volume in Scenario 1 is lower than that in Scenario 2. This is consistent with the design of the objective function, where Scenario 2 places a higher weight on oil production relative to gas. It is also noteworthy that the scheduling order differs between the two scenarios. In Scenario 2, gas opportunities were pushed toward the end of the schedule to prioritize early drilling of oil-rich targets, thereby maximizing the objective function based on discounted oil volume.

In Scenario 3, only 16 unique opportunities were scheduled. This reduction is due to the application of a risk constraint, which limits the inclusion of sidetracks to those with a total risk score of 16 or lower. As a result, all higher-risk opportunities were excluded from the optimization and alternative low risk options were added as low risk options.

Conclusions

An innovative solution to address the business challenges faced by a client operating on the Norwegian Continental Shelf has been developed and rigorously tested over the past two years. The development team sought to harness advanced digital workflows to identify optimal sidetrack opportunities and extend the operational lifespan of mature fields. This initiative was driven by the need to counteract declining production rates and manage the aging infrastructure effectively.

Adopting a collaborative and interdisciplinary approach, the team conducted comprehensive cross-domain evaluations to assess drilling sidetrack opportunities and mitigate risks associated with field development. By leveraging digital capabilities and integrating data from multiple technical domains, the project aimed to optimize drilling schedules and enhance resource efficiency. Utilizing a portfolio-based methodology and evaluating a range of scenarios, the team was successful in identifying optimized alternative drilling strategies for assets in the North Sea.

The innovative workflow is specifically designed to enhance the identification and prioritization of sidetrack opportunities in mature fields. It addresses significant challenges, including the manual handling of a large volume of sidetrack options (often exceeding 1,000 per field), the limited time and resources to study and evaluate such a large number of opportunities and the lack of an integrated collaboration platform between drilling and reservoir engineering disciplines. These challenges make it impossible to assess more than a handful of sidetrack opportunities.

The newly established workflow unlocks high quality decision making by allowing orders or magnitude more side track opportunities to be generated, evaluated and scheduled with confidence in the outcome. It significantly reduces the well planning timeline from several months to just a few weeks. Moreover, it ensures high-quality decision-making throughout each phase of the process, thereby securing the alignment of well-slot usage with the most appropriate targets and timing. The outcome is a long-term, risk-informed drilling schedule designed to maximize production and overall asset value.

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